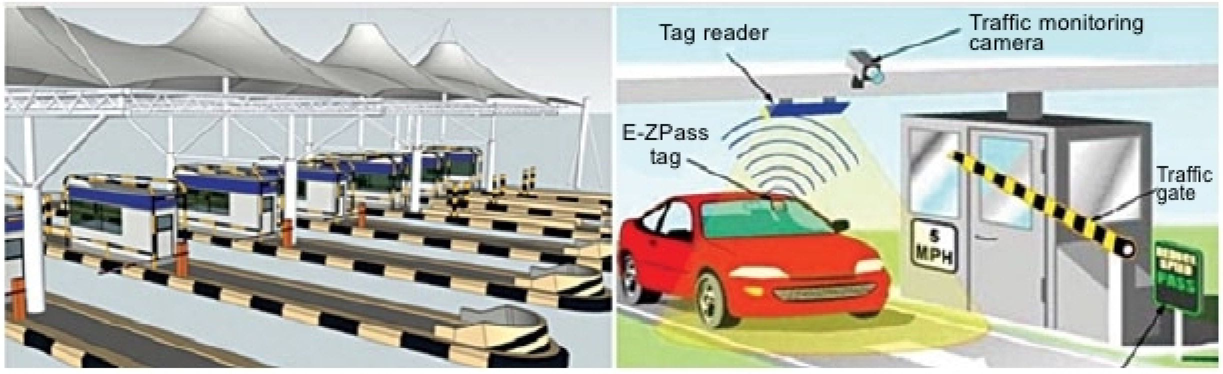

RFID-based automated vehicle identification (Credit: www.hopelandrfid.com)

of the cellphones. By measuring and analysing network data using triangulation, pattern matching or cell-sector statistics, this data can be converted into traffic flow information.

The advantage of this method is that no additional infrastructure needs to be built along the road as the existing cellphone networks are used. However, the method could be complicated in areas where the same cellphone tower serves two or more parallel routes.

Vehicle re-identification. This method requires sets of detectors mounted along the road. In this technique, a unique serial number on the number plate of the vehicle is detected at one location and then detected again (re-identified) further down the road. Travel time and speed are calculated by comparing the time at which the vehicle is detected by the pairs of sensors. This can be done using Bluetooth, RFID or serial numbers from electronic toll collection (ETC) transponders (also called toll tags).

GPS-based method. An increasing number of vehicles are now equipped with in-vehicle GPS or satnav (satellite navigation) systems that have two-way communication with a traffic data provider. Different position readings from these vehicles are used to compute vehicle speeds.

Smartphone-based monitoring. Smartphones with sensors are used to track traffic speed and density. Accelerometer data from smartphones used by car drivers is monitored to find out the traffic speed and road quality. Audio data and GPS tagging of smartphones enable identification of traffic density and possible traffic jams.

Bluetooth detection. Sensing devices implanted along the road are used to detect Bluetooth devices present in passing vehicles. By interconnecting these sensors, it is possible to calculate travel time and provide data for origin and destination matrices. This is an accurate and inexpensive way to measure travel time, and make origin and destination analysis.

Compared to other traffic measurement technologies, Bluetooth measurements have the advantages of being more accurate and non-intrusive. This leads to lower-cost installations for both permanent and temporary sites, which are generally quick to set up with little calibration requirements. In this method, counting and other applications are limited by the number of Bluetooth devices broadcasting in a vehicle.

Uses and applications

Automatic road enforcement. ITMS provides an effective traffic-enforcement system to detect and identify vehicles disobeying a speed limit or some other legal requirement. It automatically tickets the offenders based on the number plate. Traffic tickets are then sent by e-mail to the owners. The system includes:

- Speed cameras to identify over-speeding vehicles. Many systems also use radar or electromagnetic loops buried in each lane of the road to detect vehicular speed.
- Traffic light cameras to detect vehicles that cross a stop line while a red traffic light is showing.
- Bus lane cameras to identify vehicles travelling in lanes reserved for buses.
- Double white line cameras that identify vehicles crossing these lines.
- High-occupancy vehicle (HOV) lane cameras to identify vehicles violating HOV requirements.

Dynamic traffic light sequencing. Using RFID for dynamic traffic light